

Mingxuan Jin

PhD Researcher | Software Development and Interactive Systems

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EDUCATION

Cardiff University - PhD, Computer Science	Apr 2024 - present
School of Computational and Mathematical Sciences; expected completion Mar 2028	
University of Nottingham	Sep 2021 - Sep 2022
Master's, Human-Computer Interaction (HCI) (Distinction)	
Beijing University of Technology & University College Dublin	Sep 2016 - Jul 2020
Bachelor's, Software Engineering	

INDUSTRY EXPERIENCE

Matrixtime Robotics Co., Ltd.	Mar 2023 - Jul 2023
<i>Deep Learning Technology Engineer Intern Shanghai, China</i>	
- Built a Python/PyTorch computer vision pipeline for automotive defect inspection, improving mAP by 15% and reducing false rejections from 8% to below 2%. - Deployed models with TensorRT FP16, reducing end-to-end latency per 1080p image from 65 ms to 18 ms for production-line use.	
China CITIC Bank	Jul 2018 - Aug 2018
<i>Banking Intern Client Services and Data Analysis Beijing, China</i>	
Automated banking-data cleaning and reporting with Python/pandas and Excel VBA, improving weekly reporting efficiency by around 30%. Flagged large related-party transfers in tens of thousands of transactions.	

SELECTED RESEARCH AND ENGINEERING

AR Walking System with IMU Integration and AI Feedback Validation	
Developer Magic Leap 2, Python	Mar 2026 - Sep 2026
<i>ACM Conference on Human Factors in Computing Systems 2027 (under review)</i>	
M. Zhou, Mingxuan Jin, K. Stawarz, X. Peng, A. Denisova, H. Egami, T. Wen, Y. Gu, D. J. Finnegan, S. C. Barathi.	
- Built a Python IMU-data pipeline and Random Forest movement classifier for lower-limb exercises. Implemented distance-based AR guide behaviour. - Integrated a language-model API with structured inputs. Validated feedback against fixed progression decisions and implemented deterministic fallback responses for inconsistent outputs.	
VR Cycling System with Gesture Input and Synchronised Logging	
Lead developer and evaluator Unity/C#	Mar 2026 - Sep 2026
<i>ACM Conference on Human Factors in Computing Systems 2027 (under review)</i>	
Mingxuan Jin, M. Zhou, X. Peng, D. J. Finnegan, L. D. Turner, S. C. Barathi.	
Integrated exercise-bike input with gesture-based dialogue. Implemented four controlled event sequences and synchronised power and event logging; led evaluation in studies with 16 and 68 children.	
Configurable VR Cycling Platform with Branching Dialogue	
Lead developer and evaluator Unity/C#	May 2025 - Sep 2025
<i>International Journal of Human-Computer Studies (https://doi.org/10.1016/j.ijhcs.2026.103954)</i>	
M. Zhou, Mingxuan Jin (co-first author), X. Peng, K. Stawarz, A. Denisova, J. Huang, S. C. Barathi.	
Built bike-controlled navigation, gesture-based branching dialogue and six experimental configurations. Led deployment and evaluation in two studies ($n = 15$ and $n = 36$).	

SELECTED SOFTWARE PROJECTS

NLP Review Classification and Editing Web App	May 2022 - Sep 2022
<i>Python, JavaScript, HTML/CSS</i>	
Combined Naive Bayes-derived aspect features with an SVM rating classifier. Built rule-based editing suggestions and a web interface, then iterated through usability testing.	
Bot Detection Web App with Model Explanations	Feb 2020 - Jun 2020
<i>Python, Django, JavaScript</i>	
Built a Django web app with a 12-feature Random Forest classifier, Bayesian hyperparameter optimisation and visual and natural-language explanations derived from decision trees.	

CORE SKILLS

Languages: Python, C#, Java, JavaScript, SQL, R
Tools: Unity, Django, PyTorch, TensorRT, numpy, pandas, scikit-learn, MySQL, Git/GitHub
Systems: Bluetooth/IMU integration, API integration, synchronised event logging
Machine Learning: Computer vision, Random Forest, SVM